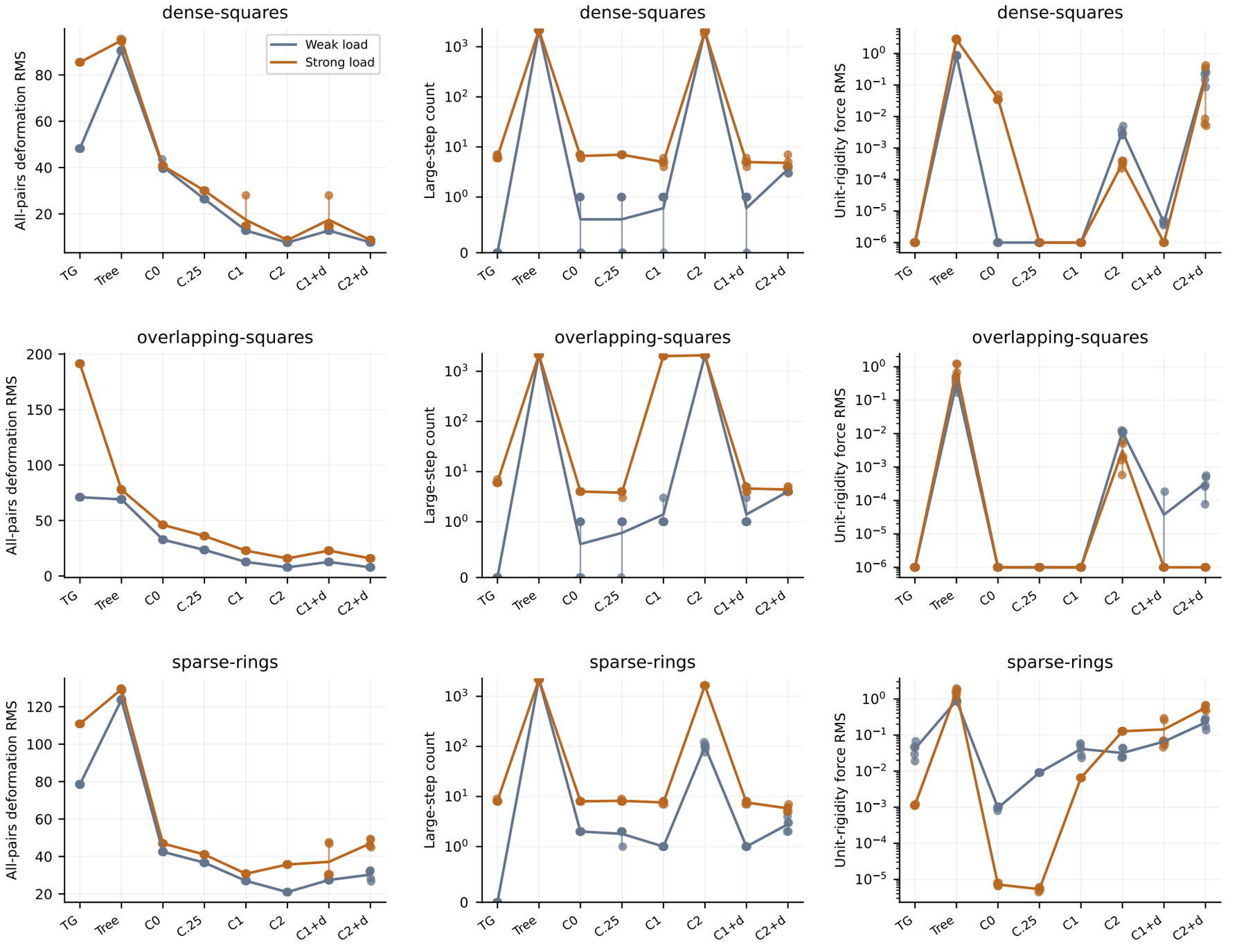
F6. Cluster rigidity, external load and dynamic cost



F6. Five matched starts per topology/load/condition. C is cluster rigidity; +d enables adaptive damping. C0 retains soft caps and compactness. Tree projection uses perimeter edges at target 120; every semantic relation remains in every simulation. All-pairs deformation uses the same unjittered shape, including unconstrained pairs. High stiffness can reduce link error while worsening sparse-shape preservation. Force values below 1e-6 are floored for display.